This is not true for every context, there are situations where we need the absolute best because even a small difference is translated in a large cost, or maybe it can help saving lives: in surgery simulations, for instance, a small error can have dire consequences, as you can imagine.

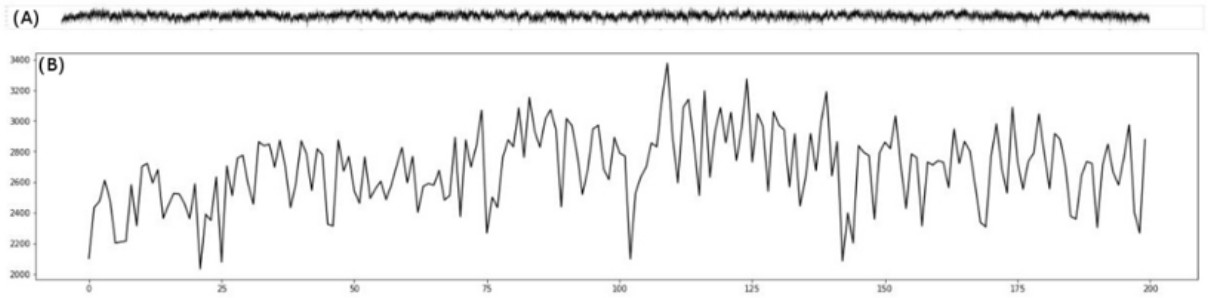


Figure 17.9 The cost landscape for the traveling salesman problem applied to the graph in figure 17.8. (A) The cost function is displayed for the whole domain: hard to make sense of it!
(B) Zooming on the first few hundreds of permutations, we can see that there are several local minima.

But since we are OK with sub-optimal solutions, this means we can use a heuristic to get an acceptable solution within a reasonable time.

Several heuristics have been developed specifically for TSP: for instance, for graphs whose distances obey the triangle inequality (like ours), a class of algorithms using the *minimum spanning tree* of a graph (see section 2.8.2) and running in *linearithmic* time, `O(n*log(n))`, can guarantee a solution whose cost is at most twice the minimum const, and on average[11] between just 15 and 20% higher than the shortest tour.